

Product Requirements Document

Clara: Agentic AI Workflow for Insurance Claims Automation

Product	Clara
Status	Hackathon MVP (working demo shipped)
Context	RWJH x Rutgers Health Hackathon - 48-hour build
Date	October 2025
Team	6-person team
Links	GitHub · Demo video

TL;DR

Manual review of high-risk insurance claims is slow, inconsistent, and dependent on a reviewer manually reading dense, unstructured medical records. Clara is an agentic AI workflow that ingests patient records and claims data, extracts the relevant medical context, and returns a structured summary plus a pass/fail recommendation with a confidence score. Clinics submit cases through a portal; insurers review Clara's analysis and accept or reject the claim. In the hackathon demo, this cut per-case review time from roughly 20 minutes to 5–7 minutes (65–75% faster) while keeping a human as the final decision-maker.

Problem & Background

Insurance claims review, particularly for high-risk or complex cases, is a manual, expert-dependent task. A reviewer has to read through unstructured medical records and claims documentation, locate the clinically relevant details, cross-reference them against policy logic, and reach a decision.

This creates three problems:

- **Time cost.** Each high-risk case takes a reviewer roughly 20 minutes of focused reading and judgment.
- **Inconsistency.** Different reviewers surface different details and weigh them differently, so outcomes vary from case to case.
- **Limited throughput.** Because the review is serial and manual, volume is capped by reviewer headcount, creating backlogs and slow decision cycles.

The underlying friction is that the information needed to make a decision is buried in unstructured text. If that context could be reliably extracted and structured, the human decision becomes faster and more consistent.

Target Users & Personas

Primary user - Insurance claims reviewer. Reviews high-risk claims and decides accept/reject. Today, they read full records manually. Clara's job is to hand them a structured summary and a recommendation so they can confirm a decision in minutes instead of reading from scratch. They need to trust the output, which means they need traceability into why Clara recommended what it did.

Submitting user - Clinic / provider staff. Submits patient cases and claims data into the system through a clinic portal. They need the submission to be simple and to require no knowledge of the AI pipeline underneath.

User Stories

- As a clinic staff member, I can submit a patient case through our portal so that it enters the review pipeline without manual handoff.
- As a claims reviewer, I receive a structured summary of the medical record so that I don't have to read the full unstructured file.
- As a claims reviewer, I see a pass/fail recommendation and a confidence score so that I can prioritize my attention and move quickly on clear-cut cases.
- As a claims reviewer, I can see the extracted medical context and flags behind a recommendation so that I can audit the reasoning before I accept or reject.
- As a claims operation, I can process many claims in parallel so that throughput isn't capped by serial manual review.

Solution Overview

Clara is an agentic pipeline with a human-in-the-loop review interface.

1. **Intake.** A clinic submits a case (patient records + claims data) through the clinic portal.
2. **Extraction.** A Python pipeline using Structify parses the unstructured medical records and pulls out relevant medical context.
3. **Decisioning.** Multiple specialized AI agents, orchestrated on the Airia Platform, extract context and produce a recommendation.
4. **Structured output.** Clara generates structured metadata for the case (e.g., medical context, claim flags), a pass/fail recommendation, and a confidence score.
5. **Human review.** The insurer reviews Clara's analysis in the review interface and accepts or rejects the claim.

The human always makes the final call. Clara accelerates and standardizes the read, it does not replace the decision.

Technical Architecture

- **Interface (frontend):** TypeScript / Next.js application providing the clinic submission portal and the insurer review interface (deployed on Vercel).
- **Extraction:** Python pipeline leveraging Structify to parse unstructured medical records.
- **Multi-agent decisioning (orchestrated on the Airia Platform):** A submitted case fans out to three specialized checks running in parallel, whose outputs feed a main synthesis model:
 - **Code validation (ICD + CPT):** checks that the diagnosis (ICD) and procedure (CPT) codes are valid and consistent with the record.
 - **Claim form completeness:** checks that the claim form is complete.
 - **Medical necessity documentation:** checks that the medical necessity of the procedure is documented.
 - **Main model (synthesis):** combines the three checks into a structured summary, a confidence score (0–1), and a pass/fail decision.
- **Output:** Structured metadata, pass/fail recommendation, and confidence score per claim, surfaced in the review interface.

Decomposing the decision into specialized checks (code validity + form completeness + medical necessity) is also what makes the recommendation traceable. A reviewer can see which check drove a flag rather than facing a single opaque score.

Success Metrics

Metric		Baseline	Target / Demo result
Per-case review time		20 min	5–7 min (65–75% reduction)
Throughput		Capped by serial manual review	Parallel processing of claims
Consistency		Reviewer-dependent	Standardized extraction across cases
Recommendation Quality		Not yet measured	To validate accuracy vs. reviewer’s final decision, with false-negative rate (legitimate claims flagged fail) tracked separately

Note: The time-reduction figures are estimates from the hackathon demo, not a validated production study.

Risks & Constraints

This is a healthcare decisioning product, so the risk surface matters as much as the feature set.

- **Wrong recommendations carry real consequences.** A false "fail" on a legitimate claim can mean denied care. This is the central reason Clara keeps a human as the final decision-maker rather than auto-adjudicating, and why the confidence score and traceability exist.
- **Trust and adoption.** Reviewers will only get the time savings if they trust the summary enough not to re-read the full record. Confidence scores and traceability into the extracted context are the levers for earning that trust.
- **Data sensitivity / compliance.** The system handles protected health information, so any production version would need to satisfy the relevant privacy and security requirements - out of scope for a hackathon build but a hard requirement for real deployment.

Scope & Future Considerations

Non-Goals - Deliberately out of scope by design:

- Clara does NOT auto-adjudicate claims. It produces a recommendation; a human insurer makes the final accept/reject decision.
- Clara is NOT a replacement for the insurer's system of record or billing systems.
- Clara does NOT handle patient-facing communication or appeals.

Future Considerations - Included to show direction, not to deliver scope:

- Integration with insurers' existing claims systems of record.
- Compliance hardening (audit logging, access controls, formal PHI handling) for production deployment.
- A feedback loop where reviewer accept/reject decisions are used to evaluate and improve the agents over time.
- Confidence-threshold routing that auto-clears the highest-confidence cases and reserves human review for the rest.